



11481 - Outer Giants around Habitable-Zone Rocky Multiple Systems

Cycle: 5, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
LHS 1140				
	1	LHS 1140 - F1140C - roll 1	MIRI Coronagraphic Imaging	(1) LHS-1140
	2	LHS 1140 - F1140C - roll 2	MIRI Coronagraphic Imaging	(1) LHS-1140
	3	LHS 1140 - F1140C - BKG	MIRI Coronagraphic Imaging	(2) LHS-1140-BACKGROUND
	4	LHS 1140 - F1140C - REF	MIRI Coronagraphic Imaging	(13) HD-4907-reference
	5	LHS 1140 - F1140C - REF - BKG	MIRI Coronagraphic Imaging	(14) HD-4907-reference-background
L 98-59				
	11	L 98-59 - F1140C - roll 1	MIRI Coronagraphic Imaging	(5) L-98-59
	12	L 98-59 - F1140C - roll 2	MIRI Coronagraphic Imaging	(5) L-98-59
	13	L 98-59 - F1140C - BKG	MIRI Coronagraphic Imaging	(6) L-98-59-BACKGROUND
	14	L 98-59 - F1140C - REF	MIRI Coronagraphic Imaging	(15) HD-74541-reference
	15	L 98-59 - F1140C - REF BKG	MIRI Coronagraphic Imaging	(16) HD-74541-reference-background

ABSTRACT

LHS 1140 and L 98-59 are nearby benchmark M dwarfs hosting compact multiple systems of transiting rocky planets, including temperate, potentially atmosphere-retaining worlds. Recalibrated Gaia DR3 astrometry combined with radial velocity and high contrast imaging data reveals unambiguous evidence for outer substellar companions in both systems, but the solutions remain degenerate in mass and semi-major axis. We propose JWST/MIRI 11.4 μm coronagraphy to (i) detect and determine the nature (giant planet vs. brown dwarf) of these companions, (ii) constrain their orbits and inclinations via joint modeling of JWST images with Gaia and archival RVs, and (iii) when detected, derive first physical properties (Teff, log g, and indicative metallicity) from mid-IR photometry. These data will deliver the first direct tests of how outer companions shape the architectures and possibly volatile histories of rocky habitable-zone planets around M dwarfs, establish two template systems for JWST/ELT follow-up.

OBSERVING DESCRIPTION

We will observe LHS 1140 and L 98-59 with the MIRI coronagraph and the F1140C filter.

Each observation is paired with a reference star, for subtracting the diffraction with the 9 points Small Grid Dither as well as a background observation.

The sequence and exposure time are defined to achieve a $S/N > 10$ at the shortest separations (0.2-0.5"), as well as to achieve the best S/N in the background limited region further away.

To that aim, we devised a specific calibration strategy, which differs from the standard procedure: the reference star and its associated background are observed for the same amount of time as the combined observations on the target (sum of the 2 rolls), noticing that MIRI coronagraphic observations once subtracted with a reference star can be limited by the background noise even at short separations. Therefore, a careful subtraction of the background is required, which is not possible with the recommended procedure because the reference star background is observed for a too small amount of time, degrading the final S/N .

The sequence is as follows:

- target roll 1 = 2 hour
- target roll 2 = 2 hour
- background target in 2 dithers = 2 x 2 hours
- reference star, repeated once = 2 x 2 hours
- background reference in 2 dithers = 2 x 2 hours

The purpose of the roll (V3PA range 7° to 14°) is to modulate the transmission of the 4QPM in the field of view in case a point source sits on one of the 4QPM transition.

That is 16 hours per target, so 32 hours of open shutter for a total charged time of 45 hours.

For each target the observations are in a non interruptible sequence for calibration purpose as well as to reduce the overheads.

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Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	LHS-1140	RA: 00 44 59.3309 (11.2472121d) Dec: -15 16 17.54 (-15.27154d) Equinox: J2000	Proper Motion RA: 318.152 mas/yr Proper Motion Dec: -596.6230000922224 mas/yr Parallax: 0.0668286999999999" Epoch of Position: 2000	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[M stars]				
	(2)	LHS-1140-BACKGROUND	RA: 00 44 55.0000 (11.2291667d) Dec: -15 16 0.00 (-15.26667d) Equinox: J2000	Proper Motion RA: 318.152 mas/yr Proper Motion Dec: -596.6230000922224 mas/yr Parallax: 0.0668286999999999" Epoch of Position: 2000	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Calibration Description=[Coronagraphic, Telescope/sky background]				
	(5)	L-98-59	RA: 08 18 7.6214 (124.5317558d) Dec: -68 18 46.81 (-68.31300d) Equinox: J2000	Proper Motion RA: 94.794 mas/yr Proper Motion Dec: -340.08400002676353 mas/yr Parallax: 0.0942664" Epoch of Position: 2000	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[M stars]				
	(6)	L-98-59-BACKGROUND	RA: 08 18 0.0000 (124.5000000d) Dec: -68 18 0.00 (-68.30000d) Equinox: J2000	Proper Motion RA: 94.794 mas/yr Proper Motion Dec: -340.08400002676353 mas/yr Parallax: 0.0942664" Epoch of Position: 2000	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Calibration Description=[Coronagraphic, Telescope/sky background]				
	(13)	HD-4907-reference	RA: 00 50 57.7518 (12.7406325d) Dec: -15 10 2.72 (-15.16742d) Equinox: J2000	Proper Motion RA: 26.843 mas/yr Proper Motion Dec: -28.53399994364736 mas/yr Parallax: 0.0022105000000000002" Epoch of Position: 2000	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Calibration Description=[Coronagraphic, Point spread function]				

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(14)	HD-4907-reference-background	RA: 00 50 51.0000 (12.7125000d) Dec: -15 10 0.00 (-15.16667d) Equinox: J2000	Parallax: 0.002210500000000002" Epoch of Position: 2000
<i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i> <i>SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.</i> <i>Category=Calibration</i> <i>Description=[Coronagraphic, Telescope/sky background]</i>			
(15)	HD-74541-reference	RA: 08 01 28.5619 (120.3690079d) Dec: -67 49 13.33 (-67.82037d) Equinox: J2000	Proper Motion RA: -3.35 mas/yr Proper Motion Dec: -15.666999934182968 mas/yr Parallax: 0.0036248" Epoch of Position: 2000
<i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i> <i>SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.</i> <i>Category=Calibration</i> <i>Description=[Coronagraphic, Point spread function]</i>			
(16)	HD-74541-reference-background	RA: 08 01 20.0000 (120.3333333d) Dec: -67 49 0.00 (-67.81667d) Equinox: J2000	Epoch of Position: 2000
<i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i> <i>SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.</i> <i>Category=Calibration</i> <i>Description=[Coronagraphic, Telescope/sky background]</i>			

Proposal 11481 - Observation 1 - Outer Giants around Habitable-Zone Rocky Multiple Systems

Observation	Proposal 11481, Observation 1: LHS 1140 - F1140C - roll 1												Fri Mar 13 21:07:04 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Coronagraphic Imaging												
	Background Observations:[LHS 1140 - F1140C - roll 2 (Obs 2), LHS 1140 - F1140C - BKG (Obs 3)]												
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
	(LHS 1140 - F1140C - roll 1 (Obs 1)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous			
	(1)	LHS-1140	RA: 00 44 59.3309 (11.2472121d) Dec: -15 16 17.54 (-15.27154d) Equinox: J2000				Proper Motion RA: 318.152 mas/yr Proper Motion Dec: -596.6230000922224 mas/yr Parallax: 0.06682869999999999" Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.												
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.												
	Category=Star Description=[M stars]												
Acquisition	#	Target	Filter	Quadrant	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID			
	1	SAME	FND	1	FAST	98	1	1	23.489	224401.9			
Template	Repeat observation												
	NO												
Dithers	#	Dither Type											
	1	NONE											
Spectral Elements	#	Coron Mask/Filter	Subarray	Mask	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	4QPM/F1140C	MASK1140	4QPM	F1140C	FASTR1	500	60	1	1	60	7204.541	224401.1

Proposal 11481 - Observation 1 - Outer Giants around Habitable-Zone Rocky Multiple Systems

PSF References	LHS 1140 - F1140C - REF (Obs 4) (PSF Reference; Filters [F1140C]) Additional Justification: false
Special Requirements	No Parallel Attachments Sequence Observations 1, 2, 3, 4, 5, Non-interruptible V3 PA Offset 2 from 1 by 7 to 14 Degrees (Same offsets in Aperture)

Proposal 11481 - Observation 2 - Outer Giants around Habitable-Zone Rocky Multiple Systems

Observation	Proposal 11481, Observation 2: LHS 1140 - F1140C - roll 2												Fri Mar 13 21:07:04 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Coronagraphic Imaging												
	Background Observations:[LHS 1140 - F1140C - roll 1 (Obs 1), LHS 1140 - F1140C - BKG (Obs 3)]												
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
	(LHS 1140 - F1140C - roll 2 (Obs 2)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous			
	(1)	LHS-1140	RA: 00 44 59.3309 (11.2472121d) Dec: -15 16 17.54 (-15.27154d) Equinox: J2000				Proper Motion RA: 318.152 mas/yr Proper Motion Dec: -596.6230000922224 mas/yr Parallax: 0.06682869999999999" Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.												
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.												
	Category=Star Description=[M stars]												
Acquisition	#	Target	Filter	Quadrant	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID			
	1	SAME	FND	1	FAST	98	1	1	23.489	224401.9			
Template	Repeat observation												
	NO												
Dithers	#	Dither Type											
	1	NONE											
Spectral Elements	#	Coron Mask/Filter	Subarray	Mask	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	4QPM/F1140C	MASK1140	4QPM	F1140C	FASTR1	500	60	1	1	60	7204.541	224401.1

Proposal 11481 - Observation 2 - Outer Giants around Habitable-Zone Rocky Multiple Systems

PSF References	LHS 1140 - F1140C - REF (Obs 4) (PSF Reference; Filters [F1140C]) Additional Justification: false
Special Requirements	No Parallel Attachments Sequence Observations 1, 2, 3, 4, 5, Non-interruptible V3 PA Offset 2 from 1 by 7 to 14 Degrees (Same offsets in Aperture)

Proposal 11481 - Observation 3 - Outer Giants around Habitable-Zone Rocky Multiple Systems

Observation	Proposal 11481, Observation 3: LHS 1140 - F1140C - BKG												Fri Mar 13 21:07:04 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Coronagraphic Imaging												
	Background Observation For: [LHS 1140 - F1140C - roll 1 (Obs 1), LHS 1140 - F1140C - roll 2 (Obs 2)]												
Diagnostics	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates					Targ. Coord. Corrections			Miscellaneous		
	(2)	LHS-1140-BACKGROUND	RA: 00 44 55.0000 (11.2291667d) Dec: -15 16 0.00 (-15.26667d) Equinox: J2000					Proper Motion RA: 318.152 mas/yr Proper Motion Dec: -596.6230000922224 mas/yr Parallax: 0.0668286999999999" Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.												
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.												
	Category=Calibration Description=[Coronagraphic, Telescope/sky background]												
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Repeat observation					Background Quadrant						
	FND	YES					1						
Dithers	#	Dither Type											
	1	BACKGROUND											
Spectral Elements	#	Coron Mask/Filter	Subarray	Mask	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	4QPM/F1140C	MASK1140	4QPM	F1140C	FASTR1	500	60	1	2	120	14409.082	224401

Proposal 11481 - Observation 3 - Outer Giants around Habitable-Zone Rocky Multiple Systems

PSF References	Additional Justification: false
Special Requirements	No Parallel Attachments Sequence Observations 1, 2, 3, 4, 5, Non-interruptible

Proposal 11481 - Observation 4 - Outer Giants around Habitable-Zone Rocky Multiple Systems

Observation	Proposal 11481, Observation 4: LHS 1140 - F1140C - REF												Fri Mar 13 21:07:04 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Coronagraphic Imaging												
	Background Observations:[LHS 1140 - F1140C - REF - BKG (Obs 5)]												
Diagnostics	(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous			
	(13)	HD-4907-reference	RA: 00 50 57.7518 (12.7406325d) Dec: -15 10 2.72 (-15.16742d) Equinox: J2000				Proper Motion RA: 26.843 mas/yr Proper Motion Dec: -28.53399994364736 mas/yr Parallax: 0.0022105000000000002" Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.												
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.												
	Category=Calibration Description=[Coronagraphic, Point spread function]												
Acquisition	#	Target	Filter	Quadrant	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID			
	1	SAME	FND	1	FAST	98	1	1	23.489	224401.11			
Template	Repeat observation												
	YES												
Dithers	#	Dither Type											
	1	9-POINT-SMALL-GRID											
Spectral Elements	#	Coron Mask/Filter	Subarray	Mask	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	4QPM/F1140C	MASK1140	4QPM	F1140C	FASTR1	500	7	1	9	63	7562.863	224401.10

Proposal 11481 - Observation 4 - Outer Giants around Habitable-Zone Rocky Multiple Systems

PSF References	PSF Reference: true
Special Requirements	No Parallel Attachments Sequence Observations 1, 2, 3, 4, 5, Non-interruptible

Proposal 11481 - Observation 5 - Outer Giants around Habitable-Zone Rocky Multiple Systems

Observation	Proposal 11481, Observation 5: LHS 1140 - F1140C - REF - BKG												Fri Mar 13 21:07:04 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Coronagraphic Imaging												
	Background Observation For: [LHS 1140 - F1140C - REF (Obs 4)]												
Diagnostics	(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous		
	(14)	HD-4907-reference-background	RA: 00 50 51.0000 (12.7125000d) Dec: -15 10 0.00 (-15.16667d) Equinox: J2000				Parallax: 0.002210500000000002" Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.												
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.												
	Category=Calibration Description=[Coronagraphic, Telescope/sky background]												
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Repeat observation				Background Quadrant							
	FND	YES				1							
Dithers	#	Dither Type											
	1	BACKGROUND											
Spectral Elements	#	Coron Mask/Filter	Subarray	Mask	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	4QPM/F1140C	MASK1140	4QPM	F1140C	FASTR1	500	64	1	2	128	15369.72	224401

Proposal 11481 - Observation 5 - Outer Giants around Habitable-Zone Rocky Multiple Systems

PSF References	Additional Justification: false
Special Requirements	No Parallel Attachments Sequence Observations 1, 2, 3, 4, 5, Non-interruptible

Proposal 11481 - Observation 11 - Outer Giants around Habitable-Zone Rocky Multiple Systems

Observation	Proposal 11481, Observation 11: L 98-59 - F1140C - roll 1												Fri Mar 13 21:07:04 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Coronagraphic Imaging												
	Background Observations:[L 98-59 - F1140C - roll 2 (Obs 12), L 98-59 - F1140C - BKG (Obs 13)]												
Diagnostics	(Visit 11:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
	(L 98-59 - F1140C - roll 1 (Obs 11)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous			
	(5)	L-98-59	RA: 08 18 7.6214 (124.5317558d) Dec: -68 18 46.81 (-68.31300d) Equinox: J2000				Proper Motion RA: 94.794 mas/yr Proper Motion Dec: -340.08400002676353 mas/yr Parallax: 0.0942664" Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.												
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.												
	Category=Star Description=[M stars]												
Acquisition	#	Target	Filter	Quadrant	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID			
	1	SAME	FND	1	FAST	66	1	1	15.819	224403.9			
Template	Repeat observation												
	NO												
Dithers	#	Dither Type											
	1	NONE											
Spectral Elements	#	Coron Mask/Filter	Subarray	Mask	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	4QPM/F1140C	MASK1140	4QPM	F1140C	FASTR1	500	60	1	1	60	7204.541	224403.1

Proposal 11481 - Observation 11 - Outer Giants around Habitable-Zone Rocky Multiple Systems

PSF References	L 98-59 - F1140C - REF (Obs 14) (PSF Reference; Filters [F1140C]) Additional Justification: false
Special Requirements	No Parallel Attachments Sequence Observations 11, 12, 13, 14, 15, Non-interruptible V3 PA Offset 12 from 11 by 7 to 14 Degrees (Same offsets in Aperture)

Proposal 11481 - Observation 12 - Outer Giants around Habitable-Zone Rocky Multiple Systems

Observation	Proposal 11481, Observation 12: L 98-59 - F1140C - roll 2												Fri Mar 13 21:07:04 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Coronagraphic Imaging												
	Background Observations:[L 98-59 - F1140C - roll 1 (Obs 11), L 98-59 - F1140C - BKG (Obs 13)]												
Diagnostics	(Visit 12:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
	(L 98-59 - F1140C - roll 2 (Obs 12)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous			
	(5)	L-98-59	RA: 08 18 7.6214 (124.5317558d) Dec: -68 18 46.81 (-68.31300d) Equinox: J2000				Proper Motion RA: 94.794 mas/yr Proper Motion Dec: -340.08400002676353 mas/yr Parallax: 0.0942664" Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.												
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.												
	Category=Star Description=[M stars]												
Acquisition	#	Target	Filter	Quadrant	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID			
	1	SAME	FND	1	FAST	66	1	1	15.819	224403.9			
Template	Repeat observation												
	NO												
Dithers	#	Dither Type											
	1	NONE											
Spectral Elements	#	Coron Mask/Filter	Subarray	Mask	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	4QPM/F1140C	MASK1140	4QPM	F1140C	FASTR1	500	60	1	1	60	7204.541	224403.1

Proposal 11481 - Observation 12 - Outer Giants around Habitable-Zone Rocky Multiple Systems

PSF References	L 98-59 - F1140C - REF (Obs 14) (PSF Reference; Filters [F1140C]) Additional Justification: false
Special Requirements	No Parallel Attachments Sequence Observations 11, 12, 13, 14, 15, Non-interruptible V3 PA Offset 12 from 11 by 7 to 14 Degrees (Same offsets in Aperture)

Proposal 11481 - Observation 13 - Outer Giants around Habitable-Zone Rocky Multiple Systems

Observation	Proposal 11481, Observation 13: L 98-59 - F1140C - BKG												Fri Mar 13 21:07:04 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Coronagraphic Imaging												
	Background Observation For: [L 98-59 - F1140C - roll 1 (Obs 11), L 98-59 - F1140C - roll 2 (Obs 12)]												
Diagnostics	(Visit 13:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates					Targ. Coord. Corrections			Miscellaneous		
	(6)	L-98-59-BACKGROUND	RA: 08 18 0.0000 (124.5000000d) Dec: -68 18 0.00 (-68.30000d) Equinox: J2000					Proper Motion RA: 94.794 mas/yr Proper Motion Dec: -340.08400002676353 mas/yr Parallax: 0.0942664" Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.												
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.												
	Category=Calibration												
	Description=[Coronagraphic, Telescope/sky background]												
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Repeat observation					Background Quadrant						
	FND	YES					1						
Dithers	#	Dither Type											
	1	BACKGROUND											
Spectral Elements	#	Coron Mask/Filter	Subarray	Mask	Filter	Readout Pattern	Groups/Int	Integrations/E xp	Exposures/Dit h	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	4QPM/F1140C	MASK1140	4QPM	F1140C	FASTR1	500	60	1	2	120	14409.082	224403

Proposal 11481 - Observation 13 - Outer Giants around Habitable-Zone Rocky Multiple Systems

PSF References	Additional Justification: false
Special Requirements	No Parallel Attachments Sequence Observations 11, 12, 13, 14, 15, Non-interruptible

Proposal 11481 - Observation 14 - Outer Giants around Habitable-Zone Rocky Multiple Systems

Observation	Proposal 11481, Observation 14: L 98-59 - F1140C - REF												Fri Mar 13 21:07:04 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Coronagraphic Imaging												
	Background Observations:[L 98-59 - F1140C - REF BKG (Obs 15)]												
Diagnostics	(Visit 14:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous			
	(15)	HD-74541-reference	RA: 08 01 28.5619 (120.3690079d) Dec: -67 49 13.33 (-67.82037d) Equinox: J2000				Proper Motion RA: -3.35 mas/yr Proper Motion Dec: -15.666999934182968 mas/yr Parallax: 0.0036248" Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.												
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Calibration Description=[Coronagraphic, Point spread function]												
Acquisition	#	Target	Filter	Quadrant	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID			
	1	SAME	FND	1	FAST	66	1	1	15.819	224403.10			
Template	Repeat observation												
	YES												
Dithers	#	Dither Type											
	1	9-POINT-SMALL-GRID											
Spectral Elements	#	Coron Mask/Filter	Subarray	Mask	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	4QPM/F1140C	MASK1140	4QPM	F1140C	FASTR1	500	7	1	9	63	7562.863	224403.11

Proposal 11481 - Observation 14 - Outer Giants around Habitable-Zone Rocky Multiple Systems

PSF References	PSF Reference: true
Special Requirements	No Parallel Attachments Sequence Observations 11, 12, 13, 14, 15, Non-interruptible

Proposal 11481 - Observation 15 - Outer Giants around Habitable-Zone Rocky Multiple Systems

Observation	Proposal 11481, Observation 15: L 98-59 - F1140C - REF BKG												Fri Mar 13 21:07:04 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Coronagraphic Imaging												
	Background Observation For: [L 98-59 - F1140C - REF (Obs 14)]												
Diagnostics	(Visit 15:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous		
	(16)	HD-74541-reference-background	RA: 08 01 20.0000 (120.3333333d) Dec: -67 49 0.00 (-67.81667d) Equinox: J2000				Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.												
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.												
	Category=Calibration Description=[Coronagraphic, Telescope/sky background]												
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Repeat observation						Background Quadrant					
	FND	YES						1					
Dithers	#	Dither Type											
	1	BACKGROUND											
Spectral Elements	#	Coron Mask/Filter	Subarray	Mask	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	4QPM/F1140C	MASK1140	4QPM	F1140C	FASTR1	500	64	1	2	128	15369.72	224403

Proposal 11481 - Observation 15 - Outer Giants around Habitable-Zone Rocky Multiple Systems

PSF References	Additional Justification: false
Special Requirements	No Parallel Attachments Sequence Observations 11, 12, 13, 14, 15, Non-interruptible